

Umit Y. Basaran

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Education

Ph.D. in Electrical Engineering, University of California, Riverside, CA 2022 – Present
Dean's Distinguished Fellowship Award | Advisor: Prof. Basak Guler

B.Sc. in Computer Science, Bilkent University, Turkiye 2017 – 2022
Full-tuition merit scholarship (top 0.01% nationwide)

Experience

Graduate Research Assistant, University of California, Riverside, CA Sep 2022 – Present
Advisor: Prof. Basak Guler

- Architected **certified machine unlearning framework** in PyTorch that eliminates dependency on retained training data, achieving **competitive forget scores** while maintaining **model accuracy** compared to **retrain-from-scratch baselines** through custom noise calibration mechanisms (**ICML 2025, CVPR 2025**).
- Engineered **secure multi-party finite-field training protocols** for neural networks and analyzed their overhead versus standard training on MNIST, Fashion-MNIST, and CIFAR-10 with up to **64 parties**, resulting in a publication in **IEEE Transactions on Information Theory (2024)**.

ML Engineer Intern, HyperbeeAI, San Francisco, CA Jun 2022 – Sep 2022

- Designed and deployed **end-to-end MLOps pipeline** with Docker and CI/CD automation that **reduced deployment time from 1 week to same-day** turnaround, enabling a **3–4x increase** in production-ready experiments for a **10–15 person** research team.
- Established Git-based development workflow (PR reviews, branch protection, automated testing) that significantly **reduced merge conflicts** and cut integration time from **2–3 days to hours** across the engineering team.

Research Intern, HyperbeeAI, San Francisco, CA Mar 2021 – Jun 2022

- Developed **neural network quantization pipeline** in PyTorch compressing YOLO, EfficientNet, ResNet, and MobileNet models from **64-bit to 1-bit** representations using quantization, pruning, and knowledge distillation, achieving superior **size–accuracy tradeoffs** and enabling deployment on resource-constrained edge devices.

ML Research Intern, ASELSAN (Turkey's largest defense contractor), Ankara, Turkiye Sep 2021 – Dec 2021

- Designed domain adaptation algorithms for **safety-critical radar systems** that improved day-to-night aircraft detection accuracy from **60–65% to 80–90%** on a **50K+ image** dataset using state-of-the-art transfer learning techniques.

Research Intern, Karel Electronics, Ankara, Turkiye Sep 2020 – Mar 2021

- Built **real-time patient monitoring system** achieving **90% detection accuracy** for MS and Parkinson's disease events using accelerometer data (x,y,z axes) and lightweight neural networks optimized for embedded deployment, validated on **5–6 patients** over a **year-long** pilot study.

Selected Technical Projects

Machine Unlearning Framework | PyTorch, Python [GitHub: [ICML'25, CVPR'25](#)]

- Pioneered **noise calibration mechanism** enabling **certified unlearning** without access to the original training data, outperforming existing methods on **forget score**, **membership inference attacks**, and **relearn time** metrics on CIFAR-10 and CIFAR-100.
- Developed a customizable neural network **linearization** method that achieves **strong convexity** during fine-tuning, enabling efficient unlearning with formal guarantees; code released for **ICML 2025** and **CVPR 2025**.

Privacy-Preserving Neural Network Training | Python, finite-field arithmetic [GitHub: [IEEE TIT'24](#)]

- Implemented **finite-field arithmetic** and **secure multi-party computation** primitives in Python with a modular API for end-to-end neural network training over Galois fields, used in the experiments of the associated **IEEE TIT 2024** paper.

Technical Skills

Programming: Python, C, C++
ML Frameworks: PyTorch, TensorFlow, JAX, NumPy
ML/CV: YOLO, EfficientNet, ResNet, MobileNet, quantization, pruning, knowledge distillation
Tools: Git, Docker, Linux, $\LaTeX$, CI/CD

Selected Publications

- **U. Y. Basaran** et al., "A certified unlearning approach without access to source data," *ICML 2025*
- **U. Y. Basaran** et al., "Towards source-free machine unlearning," *CVPR 2025*
- **X. Lu, U. Y. Basaran, and B. Guler**, "Scalable multi-round multi-party privacy-preserving neural network training," *IEEE TIT 2024*